

CERVICAL CYTOLOGY-HISTOLOGY CORRELATION QUALITY ASSURANCE REPORT

Per ASC Clinical Practice Committee Birdsong Guideline 2017
CAP Accreditation Checklist Requirement CYP.06600

Total Cases	Concordant	Major Discordant	HSIL PV+
130 cases analyzed	31 23.8%	34 26.1%	42.3% Target: >60%

CAP ALERT: Major discordance rate of 26.1% exceeds the CAP benchmark of less than 10%. Corrective action documentation required per CYP.06600.

PV+ ALERT: HSIL positive predictive value of 42.3% falls below the CAP target of 60%. Review HSIL overcall rate.

MAJOR FALSE NEGATIVES: 14 major undercall cases identified. All require mandatory slide review per Birdsong Section 8.

Executive Summary

A total of 130 cases were analyzed.

Section 1 - Concordance Summary

Classification of all cytology-histology pairs per ASC Birdsong concordance definitions.

Concordance_Class	Count	Percent
minor_discordant	65	50.0
major_discordant	34	26.15
concordant	31	23.85

Agreement Within One Grade (Birdsong Section 5)

Percentage of pairs with exact agreement or minor disagreement only. Per Birdsong, this metric is of educational interest but not statistically meaningful due to verification bias.

Metric	Count	Percentage
Agreement within one grade (concordant + minor)	96	73.8%
Exact agreement only	31	23.9%
Major discordance (outside one grade)	34	26.1%

Section 2 - Discrepancy Bucket Summary

CAP-style discrepancy classification per Birdsong Figure 1 and Figure 3. Undercall = cytology less severe than histology. Overcall = cytology more severe than histology.

Bucket	Count	Percent
MinVar	7	5.38
MajUnd	14	10.77
MinUnd	43	33.08
Agree	31	23.85
MinOver	15	11.54
MajOver	20	15.38

Section 3 - HSIL-Focused Metrics (Birdsong Section 7)

The five statistical calculations required by Birdsong Section 7 and the ASC Clinical Practice Committee. These are the primary performance metrics for cytopathology quality assurance.

Metric	Value
HSIL Pap tests with histologic HSIL (CIN2/CIN3)	42.31
HSIL Pap tests with minor discrepancies	23.08
HSIL Pap tests with major false-positive discrepancies	34.62
Number of major false-negative / cytology undercall discrepancies	14.0
PV+ for HSIL Pap tests (%)	42.31

PV+ Interpretation (Birdsong Section 7a)

LOW - Below CAP target of 60%. Review HSIL overcall rate.

Extended PV+ - High-Grade Cytology Group (HSIL + ASC-H + AGC-NEO)

Per Birdsong Figure 1, HSIL, ASC-H, and AGC-NEO cytology diagnoses are grouped together for concordance assessment. The extended PV+ reflects combined performance of the entire high-grade cytology group.

Metric	Value
Total high-grade Pap tests (HSIL + ASC-H + AGC-NEO)	61
High-grade Pap tests with HSIL+ histology	31
Extended PV+	50.8%

Section 4 - Intergrade Follow-Up Rates (Birdsong Section 8)

Per Birdsong Section 8: careful review of rates of HSIL histology following cytologic interpretations of ASC-US, ASC-H, and LSIL might reveal opportunities for improvement. Elevated rates may indicate systematic cytologic undercalling.

Cytology	Total Cases	HSIL+ Follow-up	HSIL+ Rate (%)
ASC-US	11	9	81.8
ASC-H	11	3	27.3
LSIL	17	11	64.7
AGC	14	10	71.4
AGC-NEO	9	5	55.6
AGC-EMC	4	4	100.0

Clinical interpretation: ASC-US to HSIL+ rates above 15-20% or LSIL to HSIL+ rates above 20-25% may indicate undercalling patterns requiring targeted educational slide review.

Section 5 - Key Quality Signals

Summary of key performance indicators with CAP benchmark comparisons. Status reflects comparison against published CAP and ASC thresholds.

Quality Signal	Value	CAP Threshold	Status
Major discordance rate	26.1%	< 10%	EXCEEDS
Minor discordance rate	50.0%	No threshold	-
Undercall proportion	43.9%	< 15%	EXCEEDS
Overcall proportion	26.9%	No threshold	-
HSIL PV+ (positive predictive value)	42.3%	> 60%	LOW
Extended PV+ (HSIL + ASC-H + AGC-NEO)	50.8%	> 60%	LOW
Agreement within one grade	73.8%	Educational interest	-
Major false negative count	14	Minimize	REVIEW

Section 6 - Visualizations

All figures generated by the CHC-QA Pipeline per ASC Birdsong Guideline 2017.

Figure 1 - Concordance Distribution

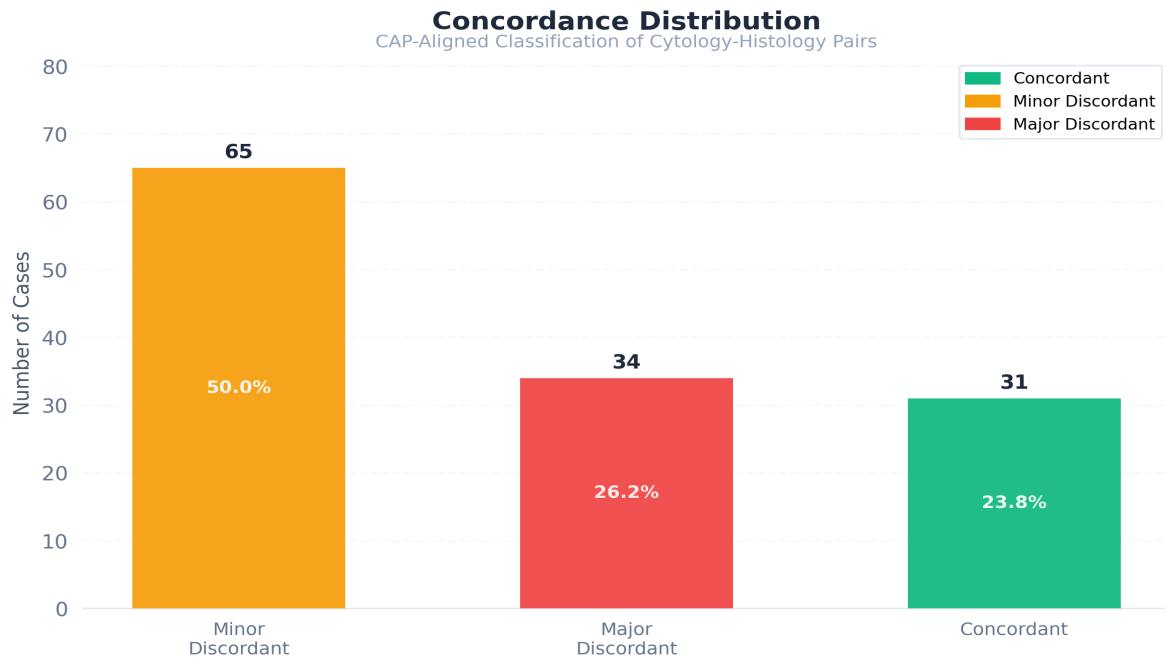


Figure 2 - Cytology-Histology Correlation Buckets

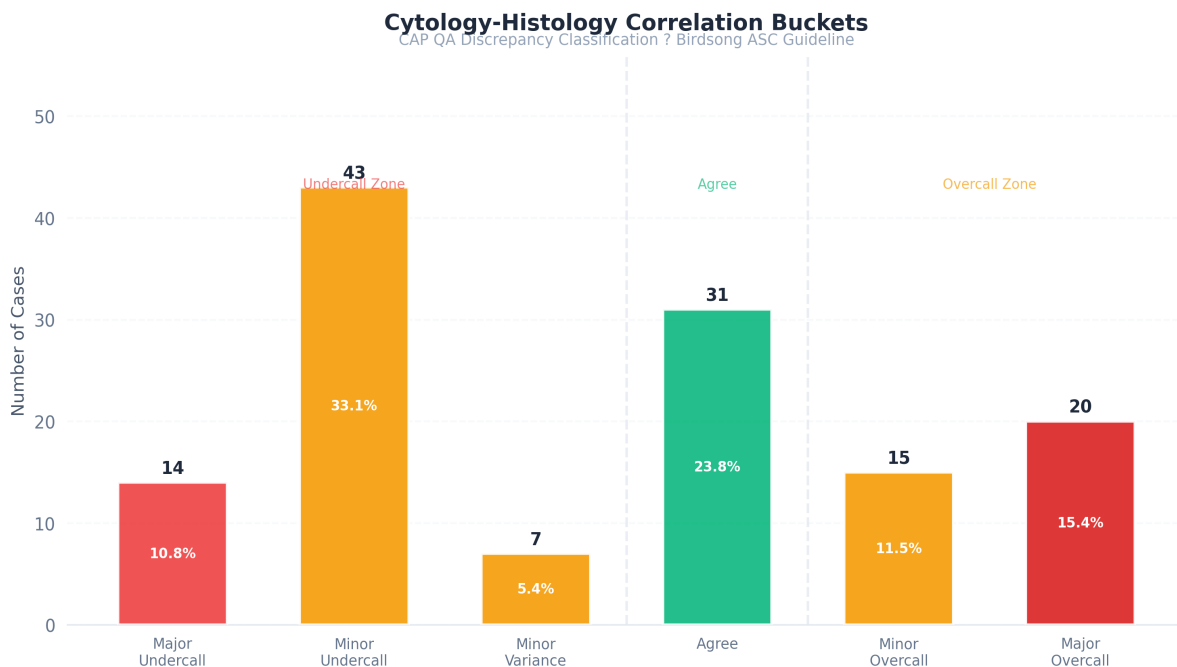


Figure 3 - HSIL Correlation Metrics

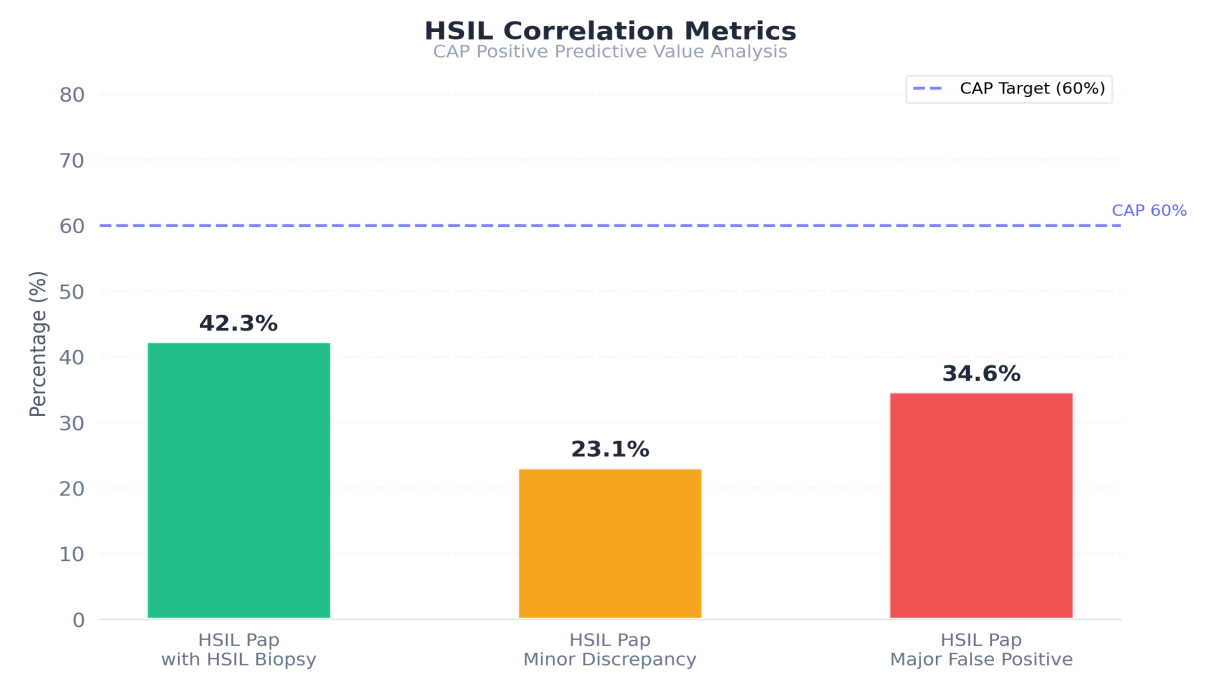


Figure 4 - Cytology vs Histology Confusion Matrix

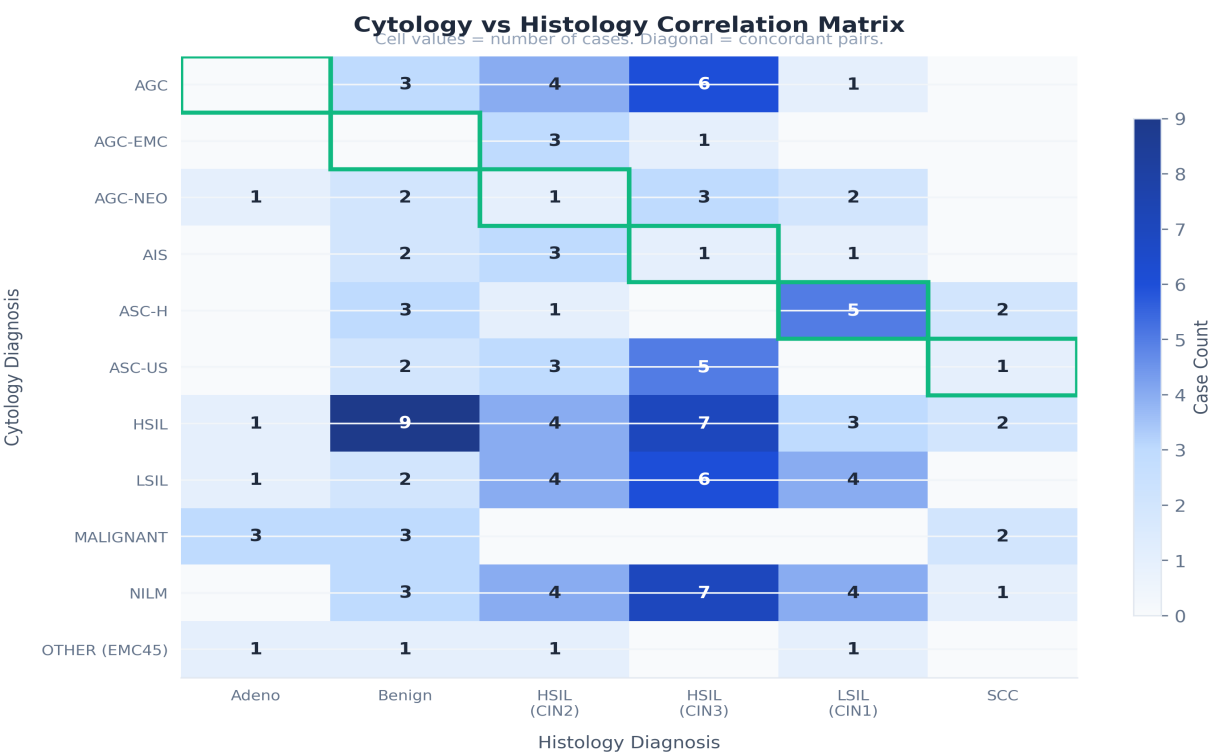
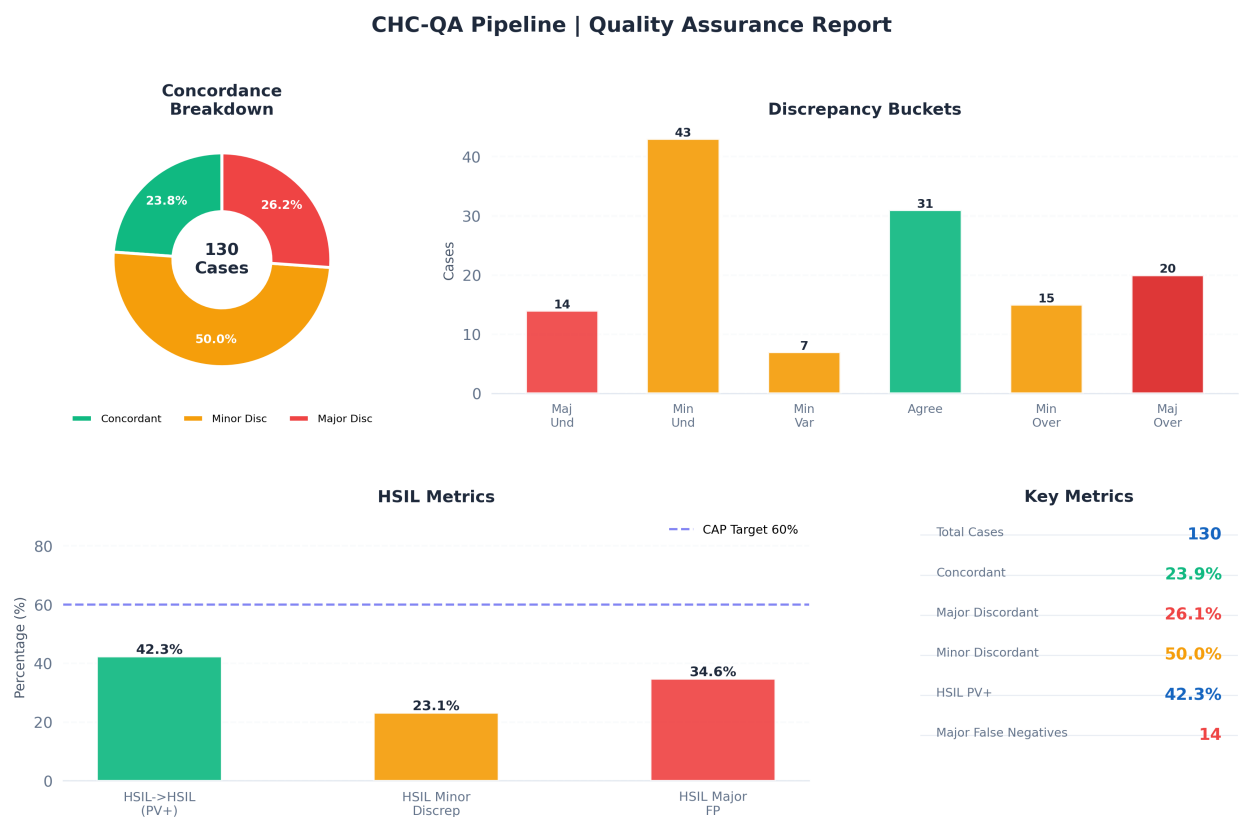


Figure 5 - Summary Dashboard



Section 7 - AI Clinical Commentary

The following commentary was generated by a locally-hosted AI language model based on the QA metrics above. This commentary should be reviewed by a qualified cytopathologist before use in official documentation.

PATTERN ANALYSIS. Analysis of 130 cytology-histology correlation pairs reveals a dominant pattern of undercalling errors. The concordance rate of 23.9% and major discordance rate of 26.1% indicate significant quality improvement opportunity requiring immediate corrective action. The undercall proportion of 43.9% and overcall proportion of 26.9% confirm undercalling as the primary performance concern.

ROOT CAUSE HYPOTHESIS. Based on Birdsong Section 8 root cause categories, the pattern of undercalling with 14 major false negative cases is most consistent with systematic screening or interpretive error. The distribution of discordance across cytology categories suggests institutional threshold issues rather than isolated incidents. Colposcopic sampling adequacy should also be evaluated as a contributing factor for overcall cases.

BENCHMARK INTERPRETATION. The HSIL PV+ of 42.3% falls below both the CAP target of 60% and the Australian Royal College standard of 65%, indicating either systematic HSIL overcalling or inadequate colposcopic tissue sampling. The major discordance rate of 26.1% exceeds the CAP benchmark of less than 10% and requires corrective action documentation per CYP.06600. Agreement within one grade of 73.8% provides additional context for overall laboratory QA performance.

INTERGRADE ANALYSIS. ASC-US to HSIL+ rate of 81.8% exceeds the expected 15-20% range ASC-H to HSIL+ rate of 27.3% is below the expected 40-60% range LSIL to HSIL+ rate of 64.7% exceeds the expected 20-25% range, suggesting systematic undercalling patterns. Per Birdsong Section 8, rates significantly outside published ranges should prompt mandatory slide review of affected cases and targeted educational intervention. Intergrade follow-up analysis is the most sensitive indicator of systematic cytologic threshold issues.

CORRECTIVE ACTION RECOMMENDATIONS. The following actions are recommended in priority order: (1) Mandatory slide review of all 14 major false negative cases by the lead cytopathologist within 14 days per Birdsong Section 8. (2) Educational session on HSIL cytomorphology for cytotechnologists. (3) Monthly CHC monitoring for 3 months to assess corrective action effectiveness. (4) Summary data presentation to all cytology staff at the next quarterly QA meeting per CAP CYP.06600 requirements.

Section 8 - Limitations

This report is generated by an automated deterministic cytology-histology correlation pipeline. The following limitations apply and should be considered when interpreting results.

1. **Verification bias:** CHC datasets are subject to verification bias due to selective histologic follow-up. Results should not be interpreted as measures of screening sensitivity or specificity.
2. **Clinical context:** Observed discrepancies should be interpreted within the context of laboratory QA review rather than as direct indicators of diagnostic error.
3. **LLM normalization:** Free-text diagnosis normalization was performed by a locally-hosted large language model. Normalized terms were validated against a controlled vocabulary. Residual normalization errors may affect a small number of cases.
4. **Pairing logic:** Cases were paired by patient identifier and date window. Complex scenarios including multiple biopsies, multiple Pap tests, and delayed follow-up may result in suboptimal pairing in a minority of cases.
5. **Root cause classification:** This pipeline classifies discordance type and direction but does not perform root cause analysis (sampling error, screening error, interpretive error). Manual review of major discordant cases is required per Birdsong Section 8.

Reference: Birdsong GG, Walker JW. Gynecologic Cytology-Histology Correlation Guideline. ASC Bulletin. 2017;LIV(2):VIII-XIII. | CAP Checklist Requirement CYP.06600 | Generated by CHC-QA Pipeline (github.com/solomoneluanu/chc-qa-pipeline)